

Per-channel autoregressive linear prediction padding in tiled CNN processing of 2D spatial data

Abstract

1 We present linear prediction as a differentiable padding method that has no train-
2 able parameters. For each channel, a stochastic autoregressive linear model is
3 fitted to the data by minimizing its noise terms in the least-squares sense. The
4 data is iteratively padded with conditional expected values of the autoregressive
5 model. We trained the convolutional RVSR super-resolution model from scratch on
6 satellite image data, using different padding methods. The simplest variant of linear
7 prediction padding reduced the mean square super-resolution error by $\sim 2\%$ at the
8 image edges, compared to zero and replication padding, with a $\sim 25\%$ increase in
9 inference time. Linear prediction padding better approximated satellite image data
10 and RVSR feature map data. With zero padding, RVSR appeared to use more of its
11 capacity to compensate for the higher approximation error. Cropping the RVSR
12 output by a few pixels reduced the super-resolution error and suppressed the impact
13 of the choice of padding method, favoring fast zero and replication padding.

14 1 Introduction

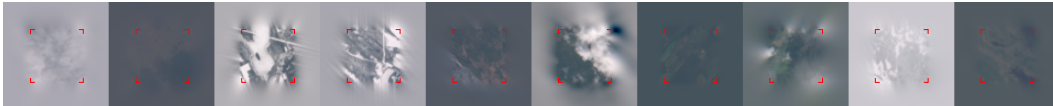


Figure 1: Satellite images padded using our linear prediction padding method (variant 1p6x7).

15 Geospatial rasters and other extensive spatial data can be processed in tiles (patches) to work around
16 memory limitations. The results are seamless if all whole-pixel shifts of the tiling grid result in
17 the same stitched results. A convolutional neural network (CNN) consisting of valid convolutions
18 and pointwise operations is equivariant to whole-pixel shifts. In such *shift equivariant* CNN-based
19 processing, each input tile must cover the receptive fields of the output pixels, i.e. input tiles must
20 be overlapped. This wastefully repeats computations. In deep CNNs, the receptive fields may be
21 thousands of pixels wide (Araujo et al. 2019), exacerbating the problem.

22 Spatial reduction (see Fig. 2) in deep CNNs is commonly compensated for by padding the input
23 of each spatial convolution, with zeros in the case of the typically used *zero padding* (zero for
24 brevity) or with the value of the nearest input pixel in *replication padding* (rep1). In the less used
25 polynomial *extrapolation padding* (extrN), a Lagrange polynomial of a degree $N - 1$ is fitted
26 to the N nearest input pixels from the same row or column, and the padding is sampled from the
27 extrapolated polynomial. extr0 is equivalent to zero and extr1 to rep1. Like these methods, our
28 *linear prediction padding* method is channel-wise, stateless and free of trainable parameters. The
29 recent *padding module* (Alrasheedi et al. [2023]) implements stateful multi-channel linear prediction
30 with coefficients trained for prediction of edge pixels of padding input using gradient-based methods.

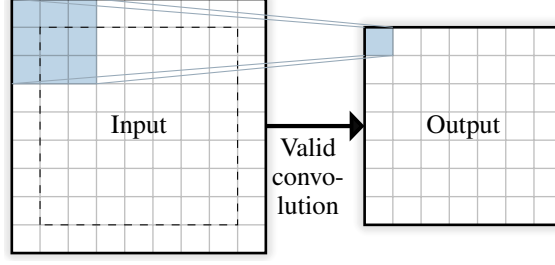


Figure 2: Valid convolution with a 3×3 kernel with origin in the middle erodes data spatially by a one-pixel-thick layer at each image edge. The receptive field of a single output pixel (■) is illustrated.

31 An ideal padding method would exactly predict the spatial data outside the padding input view. This
 32 is not possible for data coming from an effectively random process, such as natural data or a CNN
 33 feature map derived from it. Using padding increases a CNN’s error towards tile edges (Huang et al.
 34 2018). Conversely, center cropping the output of a CNN employing padding reduces the error (Huang
 35 et al. 2018). In tiled processing, the deviation from shift equivariance can be measured by the mean
 36 square deviation between overlapped CNN predictions in neighboring output tiles. The deviation is
 37 bounded from above by four times the mean square loss which can therefore be used as a proxy for it
 38 (follows from the *parallelogram law*, see appendix A). The strength of the receptive fields of output
 39 neurons typically decay super-exponentially with distance to their center (Luo et al. 2016), meaning
 40 that an output center crop discarding from each edge fewer pixels than the radius of the theoretical
 41 receptive field could reduce the prediction error close to that of a valid-convolution CNN.

42 We introduce our *linear prediction padding* (1p) method in Section 2 presenting variants based on
 43 covariance (Section 2.1, variants 1p1x1cs and 1p2x1cs, with the numbers denoting the height $\times$ width
 44 of the prediction neighborhood in vertical padding) and on autocorrelation (Section 2.2, 1p2x1, 1p2x3
 45 1p2x5, 1p3x3, 1p4x5, and 1p6x7). Implementation details are given in Section 2.3. We evaluated
 46 the performance of tiled CNN super-resolution employing different padding methods (1p variants
 47 and zero, repl, and extrN) with the super-resolution and evaluation methods given in Section 3,
 48 results in section Section 4 and our conclusions and ideas for further work in Sections 5 and 6.

49 2 Linear prediction padding method

50 Linear prediction is a method for recursively predicting data elements using nearby known or already
 51 predicted elements as input (for an introduction, see Makhoul 1975 for the 1D case and Weinlich
 52 2022 for the 2D case, both very approachable texts). Linear prediction is closely related to stochastic
 53 autoregressive (AR) processes. We can model a zero-mean version $I = J - \mu$ of 2D single-channel
 54 image or feature map data J that has a mean value of μ , by a zero-mean stationary process $\hat{I}$:

$$\hat{I}_{(y,x)+h_0} = \sum_{i=1}^P a_i \hat{I}_{(y,x)+h_i} + \varepsilon_{(y,x)+h_0}, \quad (1)$$

55 where (y, x) are integer spatial coordinates, a_i are coefficients that parameterize the process, ε is
 56 zero-mean independent and identically distributed (IID) noise, and the *extended neighborhood* h is a
 57 list of 2D coordinate offsets (relative to a shared origin), first that of the pixel of interest $\hat{I}_{(y,x)+h_0}$
 58 followed by its *neighborhood* $\hat{I}_{(y,x)+h_1 \dots P}$ in any order. Each pixel depends linearly on P neighbors
 59 (see Fig. 3). Our approach to linear prediction padding is to least-squares (LS) fit the AR model
 60 (Eq. 1) to zero-mean data I , and then to compute the padding as the expectance of the fitted AR
 61 process, assuming for conditional prediction that the data is a realization of the model, $\hat{I} = I$.

62 The LS fit is obtained by minimizing the mean square prediction error (MSE), with residual noise ε
 63 as error. Assuming real I , MSE is calculated from the error by:

$$\varepsilon_{(y,x)+h_0} = I_{(y,x)+h_0} - \sum_{i=1}^P a_i I_{(y,x)+h_i}, \quad \text{MSE} = \frac{1}{|S|} \sum_{(y,x) \in S} \left(I_{(y,x)+h_0} - \sum_{i=1}^P a_i I_{(y,x)+h_i} \right)^2, \quad (2)$$

$$S = \{(y, x) : (y, x) + h_i \in K \text{ for all } i \in [0, P]\}$$

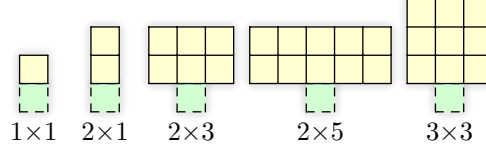


Figure 3: Illustration of some rectangular 2D neighborhoods (with pixels $\square$) next to the pixel of interest ($\blacksquare$), defining the linear dependency structure of a downwards causal AR model. The extended neighborhood 1×1 can be defined by $h = [(1, 0), (0, 0)]$ and 2×1 by $h = [(2, 0), (0, 0), (1, 0)]$.

64 where S is the set of coordinates that keeps all pixel accesses in the sums within the set of input
 65 image pixel coordinates K (see the left side of Fig. 4).

66 The noise, being zero-mean by definition, does not contribute to the AR process expected value:

$$\mathbb{E}(\hat{I}_{(y,x)+h_0}) = \mathbb{E}\left(\sum_{i=1}^P a_i \hat{I}_{(y,x)+h_i} + \varepsilon_{(y,x)+h_0}\right) = \sum_{i=1}^P a_i \mathbb{E}(\hat{I}_{(y,x)+h_i}) + 0. \quad (3)$$

67 We can recursively calculate conditional expectances (the padding) further away from known pixels
 68 (the input image), using both the known pixels $I_K = \{I_{(x,y)} : (x,y) \in K\}$ and any already
 69 calculated expectances (the nascent padding):

$$\mathbb{E}(\hat{I}_{(y,x)+h_0} | I_K) = \sum_{i=1}^P a_i \begin{cases} I_{(y,x)+h_i} & \text{if } (y,x)+h_i \in K, \\ \mathbb{E}(\hat{I}_{(y,x)+h_i} | I_K) & \text{otherwise.} \end{cases} \quad (4)$$

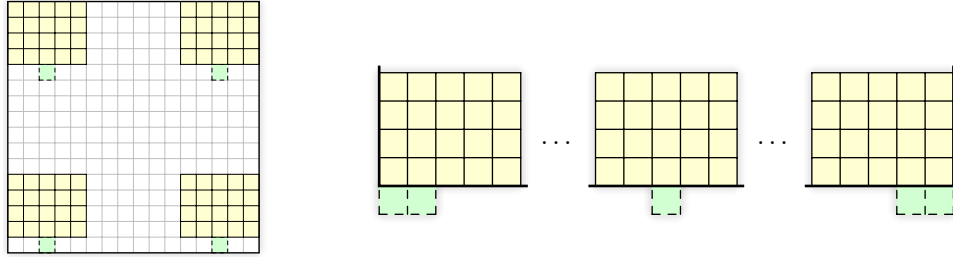


Figure 4: Downwards linear prediction padding with a 4×5 neighborhood — Left: predicted ($\blacksquare$) and neighborhood pixels ($\square$) at the corners of the rectangular area of coordinates over which MSE is calculated during fitting. Right: corner handling prevents narrowing of the recursive prediction front.

70 Our linear prediction padding process is illustrated in Fig. 5. We use rotated extended neighborhoods
 71 to pad in different directions and adjusted extended neighborhoods (see the right side of Fig. 4) to
 72 pad near the corners. We pad channels of multi-channel data separately. Before padding, we make
 73 the data zero-mean by mean subtraction, which we found necessary for numerically stable Cholesky
 74 solves of AR coefficients and to meet the assumption of a zero-mean AR process. We add the mean
 75 back after padding. Depending on the extended neighborhood shape, we use a method based on
 76 covariance or a method based on autocorrelation which is faster for larger neighborhoods.

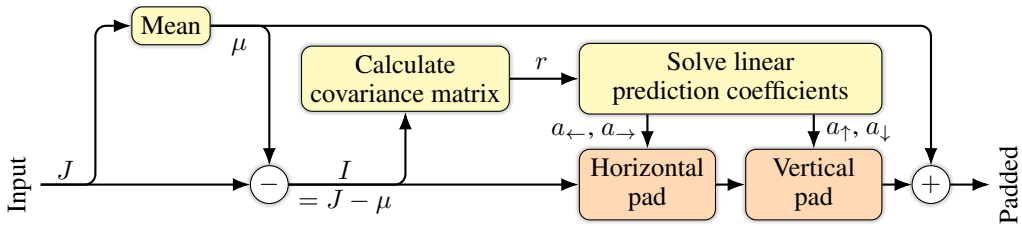


Figure 5: Linear prediction padding process for single-channel 2D data. The covariance and autocorrelation methods differ only in the *Calculate covariance matrix* block. Coefficients a are solved for centered as well as for off-center prediction for corners, for each direction indicated.

2.1 Covariance method

For one ($P = 1$) and two-pixel ($P = 2$) neighborhoods, the error to minimize can be expressed using the shorthand r_{ij} for the means of products that are elements of a *covariance matrix* r :

$$\begin{aligned} P=1: \quad \text{MSE} &= \frac{1}{|S|} \sum_{(y,x) \in S} (I_{(y,x)+h_0} - a_1 I_{(y,x)+h_1})^2 = a_1^2 r_{11} - 2a_1 r_{01} + r_{00} \\ P=2: \quad \text{MSE} &= \frac{1}{|S|} \sum_{(y,x) \in S} (I_{(y,x)+h_0} - a_1 I_{(y,x)+h_1} - a_2 I_{(y,x)+h_2})^2 \end{aligned} \quad (5)$$

$$\begin{aligned} &= a_1^2 r_{11} + 2a_1 a_2 r_{12} - 2a_1 r_{01} + a_2^2 r_{22} - 2a_2 r_{02} + r_{00}, \\ \text{where } r_{ij} &= \frac{1}{|S|} \sum_{(y,x) \in S} I_{(y,x)+h_i} I_{(y,x)+h_j}, \quad r_{ij} = r_{ji}. \end{aligned} \quad (6)$$

The LS solutions can be found by solving what are known as the *normal equations*:

$$\begin{aligned} P=1: \quad \frac{\partial \text{MSE}}{\partial a_1} &= 2a_1 r_{11} - 2r_{01} = 0 \quad \Rightarrow \quad a_1 = \frac{r_{01}}{r_{11}} \\ P=2: \quad \begin{cases} \frac{\partial \text{MSE}}{\partial a_1} = 2a_1 r_{11} + 2a_2 r_{12} - 2r_{01} = 0 \\ \frac{\partial \text{MSE}}{\partial a_2} = 2a_1 r_{12} + 2a_2 r_{22} - 2r_{02} = 0 \end{cases} &\Rightarrow \quad \begin{cases} a_1 = \frac{r_{01} r_{22} - r_{02} r_{12}}{r_{11} r_{22} - r_{12}^2} \\ a_2 = \frac{r_{02} r_{11} - r_{01} r_{12}}{r_{11} r_{22} - r_{12}^2}. \end{cases} \end{aligned} \quad (7)$$

In implementation, we used a safe version of the division operator and its derivatives that replaces infinities with zeros in results. For any P , the normal equations involve the covariance matrix r :

$$\begin{bmatrix} r_{11} & r_{12} & \dots & r_{1P} \\ r_{12} & r_{22} & \dots & r_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1P} & r_{2P} & \dots & r_{PP} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} r_{01} \\ r_{02} \\ \vdots \\ r_{0P} \end{bmatrix}. \quad (8)$$

The approach is known as the *covariance method*. We implemented it only for the neighborhoods 1×1 (method `1p1x1cs` where `cs` stands for covariance, stabilized) and 2×1 (`1p2x1cs`), using Eq. 7 and stabilization of the effectively 1D linear predictors by reciprocating the magnitude of each below-unity-magnitude root of the AR process characteristic polynomial ($1 - a_1 B$ for the 1×1 neighborhood and $1 - a_1 B - a_2 B^2$ for 2×2) of lag operator B and by obtaining the coefficients from the expanded manipulated polynomial (see Appendix B for details).

2.2 Autocorrelation method with Tukey window and zero padding

For methods `1p2x1`, `1p2x3`, `1p2x5`, `1p3x3`, `1p4x5`, and `1p6x7`, we redefined r_{ij} as normalized (N_y, N_x) -periodic autocorrelation:

$$r_{ij} = \frac{R_{h_i - h_j}}{N_y N_x}, \quad R_{(\Delta y, \Delta x)} = \sum_{y=0}^{N_y-1} \sum_{x=0}^{N_x-1} I_{(y,x)} I_{((y-\Delta y) \bmod N_y, (x-\Delta x) \bmod N_x)}. \quad (9)$$

To reduce periodization artifacts, for use in Eq. 9, we multiplied the zero-mean image horizontally and vertically by a Tukey window with a constant segment length of 50% and zero padded it sufficiently to prevent wraparound. For `1p2x5`, `1p3x3`, `1p4x5`, and `1p6x7` we accelerated calculation of R using fast Fourier transforms (FFTs) and the Wiener–Khinchin theorem, $R = \text{IDFT2}(|\text{DFT2}(I)|^2)$ for our purposes, where `DFT2` and `IDFT2` are the 2D discrete Fourier transform and its inverse.

2.3 Implementation

We implemented linear prediction padding in the JAX framework (Bradbury et al. 2025) and solved Eq. 8 using a differentiable Cholesky solver from Lineax (Rader et al. 2023), stabilizing solves by adding a small constant 10^{-7} to diagonal elements of the covariance matrix, i.e. by ridge regression.

While not dictated by the theory, we constrained our implementation to rectangular neighborhoods (as in Fig. 3) with the predicted pixel located adjacent to and centered on the neighborhood with the exception of corner padding (see Fig. 4). Our implementation benefited from the capability

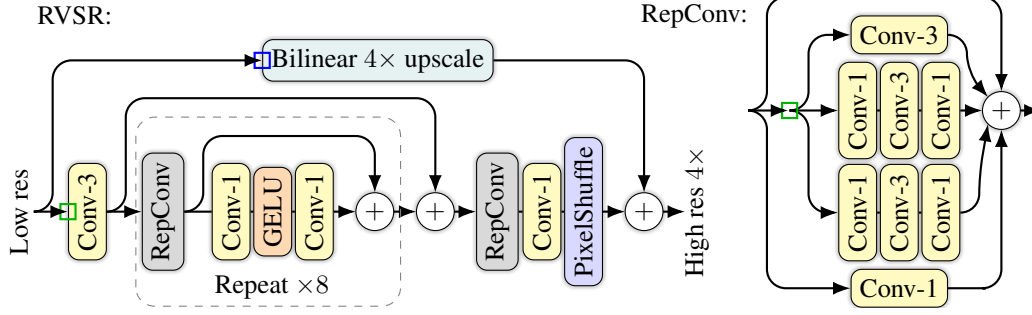


Figure 6: The convolutional RVSR super-resolution model. The spatial sizes of convolution kernels were $N \times N$ for Conv- N . Bilinear image upscaling methods in JAX and PyTorch implicitly replication pad their inputs. We padded upscale input explicitly (□) with the method configurable separately from padding of Conv inputs (□). The RepConv block was converted to a single Conv-3 for inference.

of JAX to 1) *scan* the recursion for paddings larger than a 1-pixel layer, 2) to *vmap* (vectorizing map) for parallelization of the padding front and for calculations over axes of rectangular unions of extended neighborhoods including corner handling variants, and 3) to fuse convolution with covariance statistics collection. As far as the authors are aware, such automated fusing would not take place in PyTorch that at present only offers an optimized computational kernel for zero padding.

3 Evaluation in RVSR super-resolution

We reimplemented the convolutional RVSR $4\times$ super-resolution model (Conde et al. 2024) in JAX-based Equinox (Kidger and Garcia 2021) with a fully configurable padding method (see Fig. 6)) and trained its 218 928 training-time parameters from scratch using MSE loss. For some experiments, to emulate network output center cropping, we omitted padding from the inputs of a number of the last conv layers and cropped the bilinear upscale. We define *output crop* as the number of 4-pixels-thick *shells* discarded in center cropping (see Fig. 7). We used the same padding method in the convolutional and upscale paths, with the exception of *zero-repl* and *zero-zero* where the latter designator signifies the upscale padding method. The upscale output was cropped identically to output crop, rendering *zero-repl* and *zero-zero* equivalent (denoted *zero*) for output crop ≥ 1 .

We used a dataset of 10k 512×512 pixel Sentinel 2 Level-1C RGB images (et al. 2024). We linearly mapped reflectances from $[0, 1]$ to $[-1, 1]$. We split the data into a training set of 9k images and a test set of 1k images. The mean over images and channels of the variance of the test set was 0.44 with RGB means -0.28, -0.31, and -0.22. To assemble a training batch we randomly picked images without replacement from the set, randomly cropped each image to 200×200 , created a low-resolution version by bilinearly downscaling with anti-aliasing to 50×50 , and center cropped the images to remove edge effects resulting in 192×192 target images and 48×48 input images.

We used a batch size of 64 and the Adam optimizer (Kingma 2014) with $\varepsilon = 10^{-3}$ (increased to improve stability) and default $b_1 = 0.9$ and $b_2 = 0.999$, and the learning rate linearly ramped from 5×10^{-6} to 0.014 over steps 0 to 100 (warmup, tuned for a low failure rate without sacrificing learning rate much) and from 0.014 to 0 over steps 1M to 1.5M (cooldown). We repeated the training with up to 12 different random number generator seeds.

For trained models, we also evaluated test MSE separately for shells #0–10, including in shell #10 also the rest of the shift-equivariantly processed output center. We report bootstrapped 95% confidence intervals over seeds for mean MSE and mean relative MSE difference. Test loss was calculated using center-cropped dataset images during training and by cropping at each corner in final evaluation.

Each training run took ~ 4 days on a single NVIDIA V100 32G GPU, ~ 2.5 GPU years in total consuming ~ 5 MWh of 100% renewable energy. Development and testing consumed $\sim 15\%$ additional compute. We used a V100S 32G to evaluate final MSEs, and an RTX 4070 Ti 16G for maximum batch size binary search and GPU throughput measurement at maximum batch size.

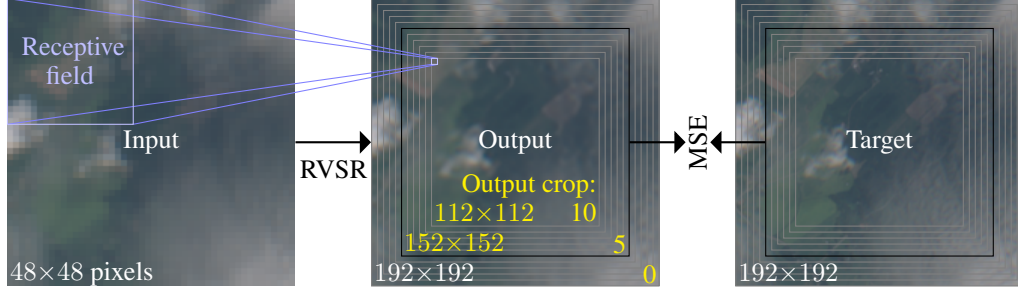


Figure 7: Illustration of RVSR super-resolution output cropping in MSE calculation, showing output crop 5 as an example. At output crop 10, the theoretical receptive fields do not include any padding.

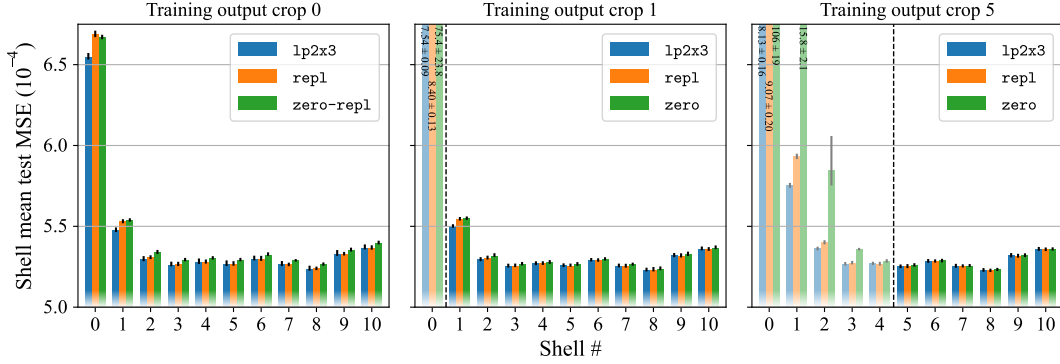


Figure 8: Super-resolution mean test MSE for select models, calculated separately for shells of equivalent width of 1 input pixel (bounded by squares in Fig. 7). Error bars indicate 95% confidence interval of the mean. Results for each output crop only include those seeds for which training was successful for every method listed. Pale-colored bars indicate exclusion of shells from training loss.

4 Results and discussion

For sample images padded using variants of 1p and other methods, see Appendix D. Super-resolution evaluation results for trained models can be found in Table 1 and Figs. 8, 9, and 10. For sample super-resolution images and a visualization of the deviation from shift equivariance see Appendix G.

In RVSR super-resolution training, we observed catastrophic Adam optimizer instability with some seed–method combinations (see Appendix F). We believe that this was a chaotic effect due to both a high learning rate and to numerical issues exemplified by differences between the equivalent `repl` and `extr1`, and not indicative of an inherent difference in training stability between the various padding methods. Overall training dynamics (Appendix C) were similar between the padding methods, with the exception of lower early-training test MSE for `repl` and the 1p methods, compared to a `zero-repl` baseline, when trained without output crop.

The different 1p methods yielded very similar super-resolution test MSEs, bringing the light-weight 1p1x1cs and 1p2x1cs to the inference throughput–MSE Pareto front (see Fig. 10) for each output crop 0 and 1. The 1p methods that used FFT-accelerated autocorrelation reached larger batch sizes in training but were limited to smaller batch sizes in inference, in comparison to 1p methods that calculated autocorrelation directly and used a similar neighborhood size.

For output crop 0, every tested 1p method yielded a 0.20–0.9% (at 95% confidence) lower mean test MSE and 1.5–2.0% lower outermost shell mean test MSE compared to the standard `zero-repl` baseline (see Tab. 1). For the other shells (see Fig. 8), 1p2x3 and `repl` were tied but consistently better than the baseline. Compared to the baseline, `repl` improved the test mean MSE by 0.1–0.7% but, notably, was 0.1–0.5% worse at the outermost shell. We hypothesize that the more clear edge signal and worse data approximation by `zero-repl`, compared to `repl`, enables and forces the network to use some of its capacity to improve shell #0 performance at the cost of overall performance. As an approximator of the internal feature maps of a trained RVSR network (with GELU activation),

Output crop	Conv and upscale padding method(s)	FFT ($\mathcal{F}$) or direct (D) autocorrelation	Maximum training batch size (images) $\uparrow$	Training throughput (images/s) $\uparrow$	Maximum inference batch size (images) $\uparrow$	Inference throughput (10^6 pixels/s) $\uparrow$	Mean test MSE (10^{-6}) $\downarrow$	Mean test MSE diff to zero-repl (%) $\downarrow$	Outermost shell mean test MSE diff to zero-repl (%) $\downarrow$
0	extr1		266	480	6741	865	545.0 ± 2.4	-0.25 ± 0.31	0.43 ± 0.24
	extr2		266	480	6741	759	550.5 ± 2.9	0.77 ± 0.35	5.67 ± 0.53
	extr3		266	478	6741	758	559.6 ± 1.9	2.38 ± 0.39	11.98 ± 0.40
	lp1x1cs		237	464	6741	722	543.3 ± 3.3	-0.64 ± 0.39	-1.82 ± 0.28
	lp2x1	D	190	433	6741	498	543.2 ± 2.3	-0.63 ± 0.25	-1.71 ± 0.19
	lp2x1cs		189	442	6741	632	542.9 ± 2.9	-0.68 ± 0.28	-1.88 ± 0.14
	lp2x3	D	187	429	6741	480	542.8 ± 3.0	-0.71 ± 0.19	-1.90 ± 0.18
	lp2x5	$\mathcal{F}$	240	453	6020	382	542.7 ± 2.5	-0.72 ± 0.17	-1.90 ± 0.11
	lp3x3	$\mathcal{F}$	241	455	6020	394	543.2 ± 2.5	-0.65 ± 0.25	-1.89 ± 0.12
	lp4x5	$\mathcal{F}$	236	447	6020	356	543.2 ± 3.5	-0.63 ± 0.39	-1.77 ± 0.18
	lp6x7	$\mathcal{F}$	191	432	6020	266	543.7 ± 1.8	-0.49 ± 0.23	-1.34 ± 0.23
	repl		266	481	6880	870	544.6 ± 3.1	-0.39 ± 0.30	0.28 ± 0.17
	zero-repl		263	472	6880	964	546.6 ± 1.9	0.00	0.00
	zero-zero		263	472	6741	960	547.1 ± 1.8	0.09 ± 0.26	1.32 ± 0.15
1	lp1x1cs		238	468	7314	695	532.2 ± 1.6	-0.27 ± 0.29	-0.93 ± 0.16
	lp2x1cs		191	444	7314	608	532.0 ± 1.8	-0.27 ± 0.17	-0.88 ± 0.20
	lp2x3	D	188	436	7314	473	532.5 ± 2.3	-0.22 ± 0.26	-0.88 ± 0.13
	repl		268	481	7314	808	532.8 ± 2.6	-0.16 ± 0.31	-0.07 ± 0.13
	zero		268	483	7314	896	533.6 ± 2.0	0.00	0.00
5	lp2x3	D	257	486	8403	465	531.8 ± 2.8	-0.05 ± 0.30	-0.16 ± 0.14
	repl		325	518	8403	664	531.7 ± 2.5	-0.06 ± 0.27	-0.14 ± 0.17
	zero		325	518	8403	706	532.0 ± 2.2	0.00	0.00

Table 1: RVSR evaluation results. 95% confidence intervals are reported (gray means an insufficient number of runs). The best results for each output crop are in **bold**. For output crop 0 we tested zero padding of convolution inputs (method names beginning with zero-) together with both zero and replication padding of upscale inputs (indicated after the hyphen). Overall, linear prediction methods had the lowest test MSE while repl and zero were the fastest and used the least memory.

zero has up to 40-fold larger error than repl and lp2x3 (see Fig. 11). The worse approximation by zero is illustrated by the up to 10-fold MSE error growth for shells not included in the training loss and outside the output crop (shaded gray in Fig. 8), compared to repl and lp2x3. Compared to explicit zero, the default implicit repl in bilinear upscale gave a lower outermost shell MSE.

For output crop 1, the choice of padding method mattered less, with lp2x3 improving upon the baseline at the two outermost shells. Models trained with output crop 5 saw no difference from the choice of padding method. Furthermore, the test MSEs have only relatively modest differences between output crops. At shells $\# \geq 5$, all models perform similarly with the exception of the somewhat worse output crop 0 zero-repl. This might be because training with output crop doesn't free up sufficient capacity to decrease MSE in the remaining image area for repl and lp2x3.

For $\text{extr}N$ we found test MSE to increase with N , with extr3 giving 11–12% worse outermost shell mean MSE compared to baseline. In contrast, Leng and Thiyagalingam 2023a found the equivalent of extr3 (see GitHub issue #2 in Leng and Thiyagalingam 2023b for a numerical demonstration of equivalence) to give better results in a U-net super-resolution task than zero or repl, which we suspect was due to their use of blurred inputs (Gaussian blur of standard deviation $\sigma = 3$) that are better approximated by the higher-degree extr3 method (see Appendix E).

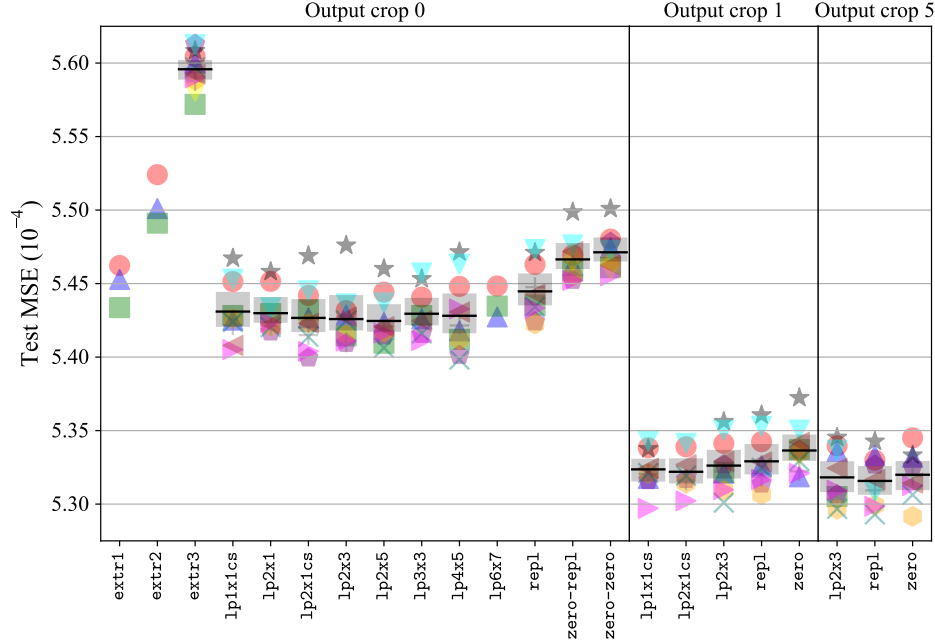


Figure 9: Super-resolution test MSE with mean (—) and the 95% confidence interval of the mean (■) across all successful training runs. For a tabular presentation and seed values, see Appendix F.

5 Conclusions

Using linear prediction padding (1p) instead of zero or replication padding (repl) improved slightly the quality of CNN-based super-resolution, in particular near image borders, at a moderate added time cost. Center-cropping the network output leveled the differences in output-target mean square error between padding methods. At output crop 5 the stitching artifacts due to deviation from shift equivariance were no longer visible.

Considering padding as autoregressive estimation of data and feature maps explains some of the differences between padding methods. However, the tested CNN architecture learned to compensate the elevated super-resolution error near the image edge to roughly the same magnitude for all the tested padding methods, including zero which has an exceptionally high estimation error. The slightly higher overall super-resolution error with zero supports the hypothesis that more network capacity is consumed by the compensation of the larger padding error.

Our results might not directly apply to other CNN architectures and tasks. Covariance statistics may suffer from the small sample problem with spatially tiny inputs such as encodings. Larger effective receptive fields may favor 1p, whereas workloads with a higher level of spatial inhomogeneity, lower spatial correlations in network input or feature maps (in particular spatially whitened data), or higher nonlinearity in spatial dependencies would likely make 1p less useful, favoring zero for its clear edge signaling. If using 1p in CNN-based processing of images with *framing*, for example photos of objects, any needed location information might need to come from another source than the padding.

Our JAX 1p padding implementation and source code for reproducing the results of this article are included in the enclosed zip file. Our lp1x1cs and lp2x1cs methods would be the most straightforward ones to port to other frameworks.

6 Further work

We have yet to explore 1) using a spatially weighted loss to level spatial differences in error, 2) using batch rather than image statistics for a larger statistical sample, 3) increasing the sample size by giving 1p memory of past statistics, 4) learning rather than solving 1p coefficients, 5) modeling dependencies between channels, 6) instead of output cropping, cross-fading adjacent output tiles and

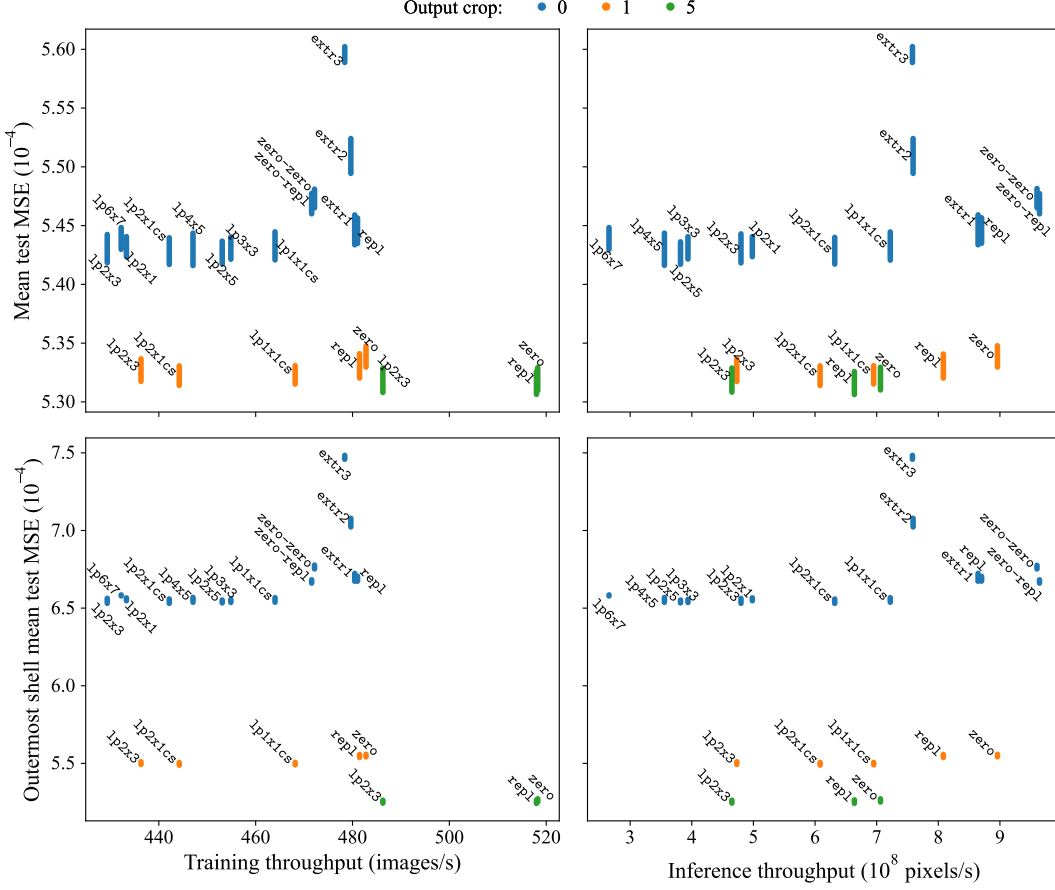


Figure 10: 95% confidence interval of the mean super-resolution MSE of each model vs. training/inference throughput evaluated using maximum training/inference batch sizes. Except for the outermost shell mean test MSE, the results do not apply to other than the used 48×48 pixel input image size because of the difference in the ratio of output pixels influenced by padding.

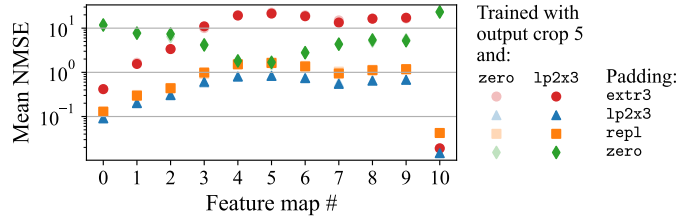


Figure 11: Test mean-over-seeds NMSE (MSE divided by data variance) of padding each convolution layer input (low-res input #0 and feature maps #1–9) and high-res network output (#10) from a 28×28 -pixel center crop to 30×30 pixels, in a trained RVSR network. Padding by mean would give $\text{NMSE} = 1$. Sparsity differences (Aimar et al. 2018) may contribute to predictability differences.

207 optimizing the cross-fade curves during training, 7) setting the padding method separately for each
 208 feature map based on its spatial autocorrelation, 8) accelerating solves by taking advantage of the
 209 covariance matrix structure, and 9) use of 1p in other CNN architectures and training settings.

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265 A A triangle-inequality-like inequality for squared distances

266 This section gives a derivation of the result that the mean square deviation between two vectors of
 267 equal length is bounded from above by four times their average mean square deviation from a third
 268 vector of equal length.

269 Let A , B , and C be length- N vectors of real numbers. Denoting the sum of squares of elements
 270 of vector x by $\|x\|^2$, the parallelogram law (see Jordan and Neumann [1935]) for $x = A - C$ and
 271 $y = C - B$ is:

$$\begin{aligned} & \|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2) \\ \iff & \|A - C + C - B\|^2 + \|A - C - C + B\|^2 = 2(\|A - C\|^2 + \|C - B\|^2) \quad (10) \\ \iff & \|A - B\|^2 + \|A - 2C + B\|^2 = 2(\|A - C\|^2 + \|C - B\|^2). \end{aligned}$$

272 By subtracting the non-negative $\|A - 2C + B\|^2$ from the left side we get an inequality similar to
 273 the triangle inequality, but for squared distances rather than distances:

$$\|A - B\|^2 \leq 2(\|A - C\|^2 + \|B - C\|^2). \quad (11)$$

274 By multiplying both sides by $\frac{1}{N}$ with N the common length of the vectors:

$$\frac{1}{N}\|A - B\|^2 \leq 2\left(\frac{1}{N}\|A - C\|^2 + \frac{1}{N}\|B - C\|^2\right), \quad (12)$$

275 and by identifying $\frac{1}{N}\|x - y\|^2$ as the mean square deviation (MSD) between x and y , we can write:

$$\begin{aligned} & \text{MSD}(A, B) \leq 2(\text{MSD}(A, C) + \text{MSD}(B, C)) \\ \iff & \text{MSD}(A, B) \leq 4 \text{mean}(\text{MSD}(A, C), \text{MSD}(B, C)). \end{aligned} \quad (13)$$

276 This means that the mean square deviation between two predictions A and B is bounded from above
 277 by four times their average mean square prediction error with C as the ground truth.

278 B Stabilization of 1D covariance method linear prediction

279 For 1D linear prediction neighborhoods 1×1 (stabilized covariance method `1p1x1cs`) and 2×1
 280 `1p2x1cs`), the padding procedure in one direction (Eq. 4) using already calculated coefficients $a_{1 \dots P}$
 281 is equivalent to a discrete-time linear time-invariant (LTI) system having a causal recursion:

$$1 \times 1 : y[k] = a_1 y[k-1] + b_0 x[k], \quad 2 \times 1 : y[k] = a_1 y[k-1] + a_2 y[k-2] + b_0 x[k], \quad (14)$$

282 where $y[k]$ are known pixel values or padding pixels, $x[k]$ are input pixels with $x[k] = 0$ for all k
 283 with an inconsequential input coefficient b_0 . The corresponding transfer functions are:

$$\begin{aligned} 1 \times 1 : \quad H(z) &= \frac{b_0}{1 - a_1 z^{-1}} \\ &= \frac{b_0 z}{z - p_0}, \\ p_0 &= a_1 \\ 2 \times 1 : \quad H(z) &= \frac{b_0}{1 - a_1 z^{-1} - a_2 z^{-2}} \\ &= \frac{b_0 z^2}{(z - p_0)(z - p_1)}, \\ p_0 &= \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2} \\ p_1 &= \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{2}, \end{aligned} \quad (15)$$

284 where b_0 is an input scaling factor, z^{-1} represents a delay of one sampling period, and $H(e^{i\omega})$,
 285 represents the frequency response of the system with ω the frequency in radians per sampling period,
 286 e the natural number, and i the imaginary unit.

287 The system is stable if all poles p of the transfer function lie inside the complex z -plane unit circle. A
 288 stationary autoregressive process is stable. If the coefficients were found by solving normal equations
 289 with approximate covariances, then stability is not guaranteed. In practice, stability is needed to
 290 prevent blow-up of the padding output when padding recursively.

291 By reciprocating the z -plane radius of all poles that have radius > 1 (i.e. $|p| > 1$), the system can be
 292 made stable, or marginally stable in case any of the poles lie at radius 1 exactly. An unstable system
 293 has no well-defined frequency response, but we can still compute $H(e^{i\omega})$. The stabilization alters the
 294 phase of $H(e^{i\omega})$ but maintains its magnitude up to a constant scaling factor that is inconsequential with
 295 zero input, thus preserving the essential power-spectral characteristics of the autoregressive process.
 296 Magnitude scaling could be compensated for by setting $b'_0 = b_0(1 - \sum_{i=0}^{i=P} a'_i)/(1 - \sum_{i=0}^{i=P} a_i)$
 297 where $'$ denotes updated variables. The choice of b_0 is inconsequential with constant zero input but
 298 would matter in *generative* padding with x a white noise *innovation*.

299 For a 2×1 neighborhood, if what's under the square root in Eq. 15 is negative, $a_1^2 + 4a_2 < 0$, then the
 300 poles are complex and form a complex conjugate pair. Otherwise, both poles are real. The squared
 301 magnitudes of the complex conjugate pair of poles are equal, $|p_0|^2 = |p_1|^2 = -a_2$. The complex
 302 poles lie outside the unit circle only if $-a_2 > 1$, in which case the system can be stabilized by
 303 $a'_1 = -a_1/a_2$, $a'_2 = 1/a_1$. Real poles p_0 and p_1 can be found using Eq. 15, they can be reciprocated
 304 when necessary, and the modified coefficients can be extracted from the expanded form of the
 305 numerator polynomial as: $a'_1 = p'_0 + p'_1$ and $a'_2 = p'_0 p'_1$.

306 The characteristic polynomial of the AR process is the denominator of the transfer function Eq. 15
 307 written with lag operator $B = z^{-1}$. With the characteristic polynomial the stability condition is that
 308 all roots are outside the unit circle. Stabilization via the characteristic polynomial would manipulate
 309 the coefficients identically to what was presented above.

310 C RVSR super-resolution training loss histories

Some RVSR super-resolution training loss histories averaged over seeds are shown in Fig. 12

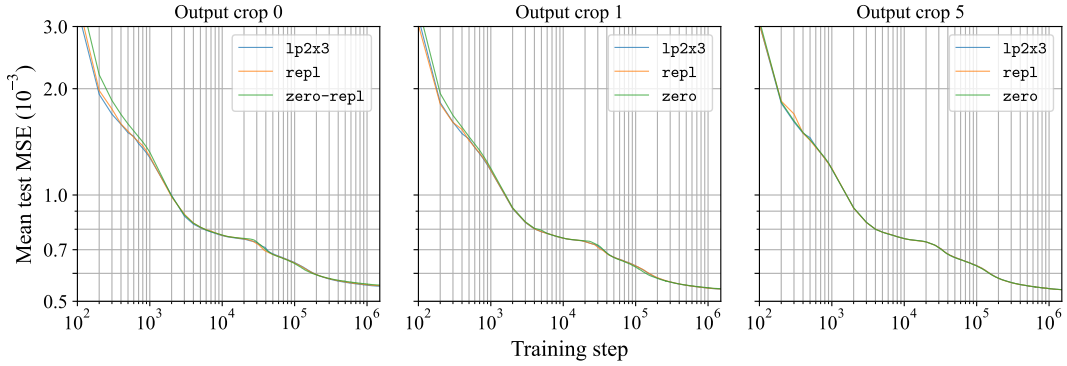


Figure 12: Super-resolution mean test loss during training for select padding methods, including for each output crop only those seeds that resulted in a successful training run for every method listed.

311

312 D Sample padded images

Fig. 13 shows sample padded images. For information about the image dataset, see section 3.



Figure 13: Non-cherry-picked 48×48 pixel test set satellite images padded with 24 pixels on each side, using different methods. The first row shows the ground truth (*target*) image from which each padding input was center-cropped (see the *zero* row for the crop boundaries). The *lp* methods with larger neighborhood widths capture directional regularities with larger slopes off the padding direction. The *extr2* and *extr3* recursions are unstable. Color channels have been clipped to reflectance range 0–0.8.

E Effect of blur on padding error

To simulate padding input data having an adjustable degree of blurriness or a rate of frequency spectral decay, we model Gaussian-blurred white-noise data by uniformly sampling a zero-mean Gaussian process x of unit variance and with Gaussian covariance as function of lag d :

$$\text{Cov}(x[i], x[i + d]) = \kappa(d) = \exp\left(-\frac{d^2}{2\sigma^2}\right) \quad \text{for all integer } i, \quad (16)$$

with σ corresponding to the standard deviation of the Gaussian blur. A general linear right padding method approximates $x[0]$ from the P nearest samples by $\hat{x}[0] = \sum_{i=1}^P a_i x[-i]$. We define the padding error ε by:

$$\varepsilon = x[0] - \hat{x}[0] = -\sum_{i=0}^P a_i x[-i], \quad a_0 = -1, \quad (17)$$

prepending $a_{1\dots P}$ with $a_0 = -1$ for convenience. The normalized mean square of the zero-mean ε is:

$$\text{NMSE} = \frac{\text{E}(\varepsilon^2)}{\text{Var}(x)} = \frac{\text{Var}(\varepsilon)}{1} = \sum_{i=0}^P \sum_{j=0}^P \text{Cov}(a_i x[-i], a_j x[-j]) = \sum_{i=0}^P \sum_{j=0}^P a_i a_j \kappa(j - i) \quad (18)$$

We compare in Fig. 14 the MSE for the following 1D padding methods with some equivalencies:

Method(s)	P	$a_{0\dots P}$
zero, extr0	0	-1
repl, extr1	1	-1, 1
extr2	2	-1, 2, -1
extr3	3	-1, 3, -3, 1
lp1x1cs	1	-1, a_1
lp2x1, lp2x1cs	2	-1, a_1, a_2
lp3x1	3	-1, a_1, a_2, a_3

with $a_{1\dots P}$ for lp1x1cs, lp2x1, and lp2x1cs from Eq. 7 and for lp3x1 from solving Eq. 8 using Levinson recursion, with known autocorrelation-like covariances $r_{ij} = \kappa(j - i)$, corresponding to the limiting case of infinitely vast padding input providing covariance statistics matching the covariances of the Gaussian process.

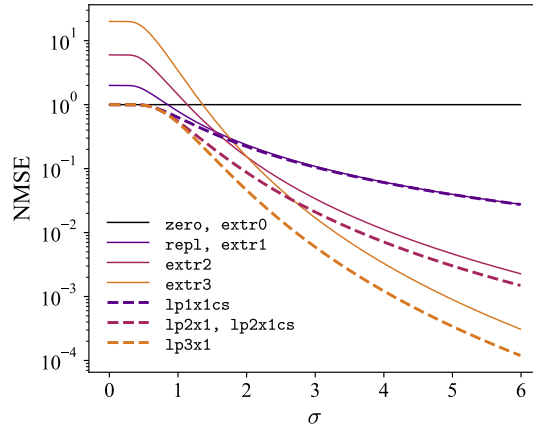


Figure 14: Theoretical padding NMSE as function of standard deviation σ of Gaussian blur applied to white-noise data. Padding input is the blurred data normalized to unit variance. Increasing the order of the extr padding mode increases the error for low σ and decreases it for high σ . Each lp method has been least-squares fit to the known signal model and thus attains the lowest possible MSE given the number of predictors. zero remains ideal for data with zero autocorrelation at non-zero lags.

327 F Trained models

output crop	padding method	seed / symbol											
		0 ●	1 ▲	2 ■	3 ◆	4 ◆	5 ●	6 ★	7 ×	8 +	9 ▼	10 ▶	11 ◀
0	lp1x1cs	✓	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓
	lp2x1	✓	✗	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	lp2x1cs	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	lp2x3	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	lp2x5	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	lp3x3	✓	✓	✓	✗	✓	✗	✓	✓	✓	✓	✓	✓
	lp4x5	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	lp6x7	✓	✓	✓	—	—	—	—	—	—	—	—	—
	zero-repl	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✓	✓
	zero-zero	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✓	✓
	repl	✓	✗	✓	✗	✓	—	✓	✓	✓	✓	✓	✓
	extr1	✓	✓	✓	—	—	—	—	—	—	—	—	—
	extr2	✓	✓	✓	—	—	—	—	—	—	—	—	—
	extr3	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1	lp1x1cs	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓
	lp2x1cs	✓	✗	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓
	lp2x3	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	zero	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	repl	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓
5	lp2x3	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
	zero	✓	✓	✗	✗	✓	✓	✓	✓	✓	✗	✓	✓
	repl	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓

329 Super-resolution training was either successful (✓), failed (✗), or skipped (—) for different combina-
 330 tions of padding methods and the random number generator seed. We report loss differences between
 331 models using only seeds that resulted in successful training for both of the compared models.

332 G Tiled processing samples and deviation from shift equivariance

333 Figs. 16–18 show non-cherry-picked stitched tiled processing results from models trained with seed
 334 10, using different output crops. Fig. 15 show the corresponding inputs and targets from the test set.
 335 Figs. 19–21 visualize deviation from shift equivariance for the same images.



Figure 15: The super-resolution inputs and targets from the test set, cropped to the same view as the sample results.

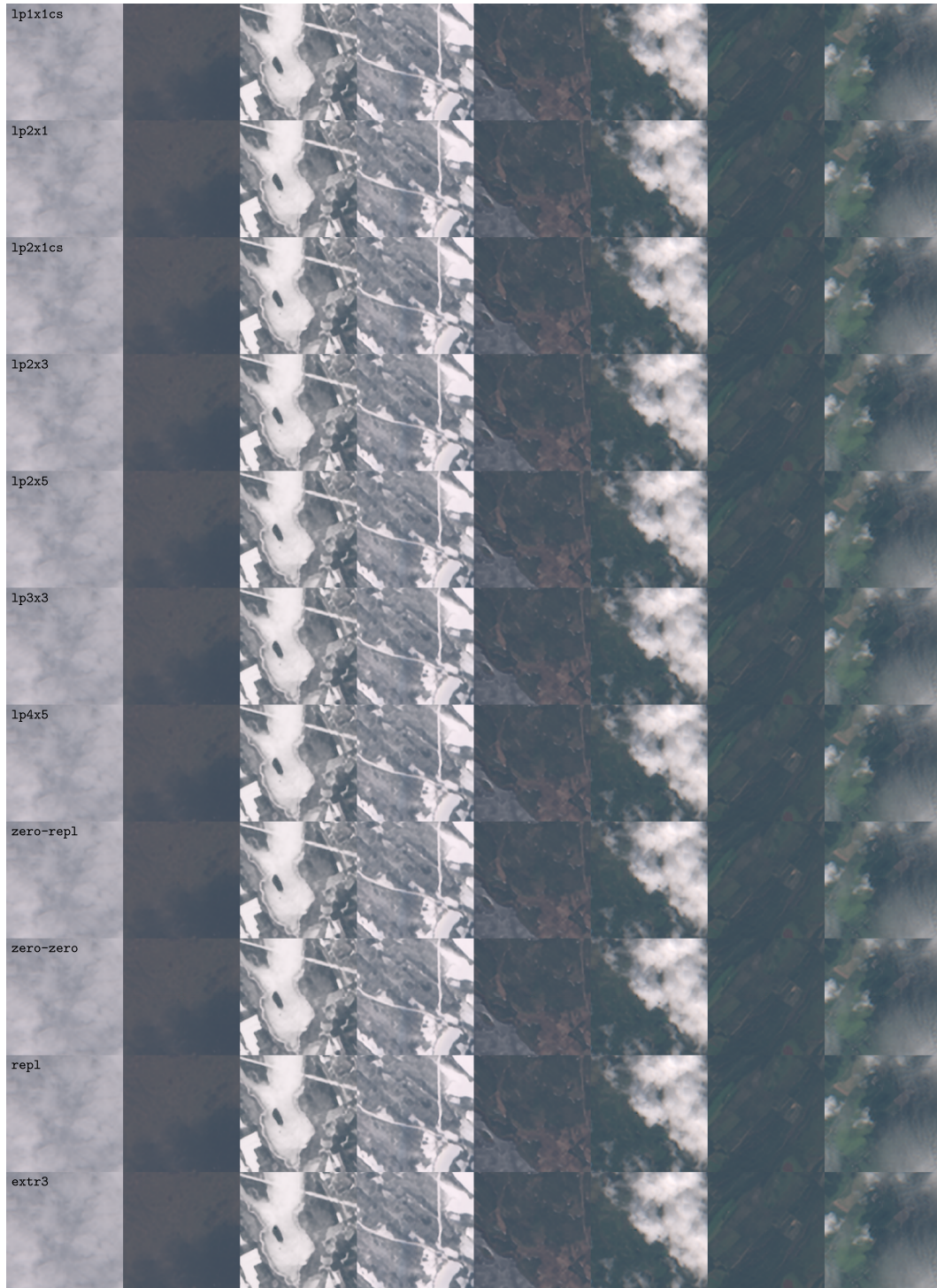


Figure 16: Stitched RVSR super-resolution results with output crop 0. Four output tiles, each as large as the images shown, were stitched together and cropped, with a corner of the tiling grid at the center of each image. A cross-hair-shaped discontinuity artifact is formed due to deviation from shift equivariance. Black represents reflectance 0 and white represents reflectance 0.8 on every channel.

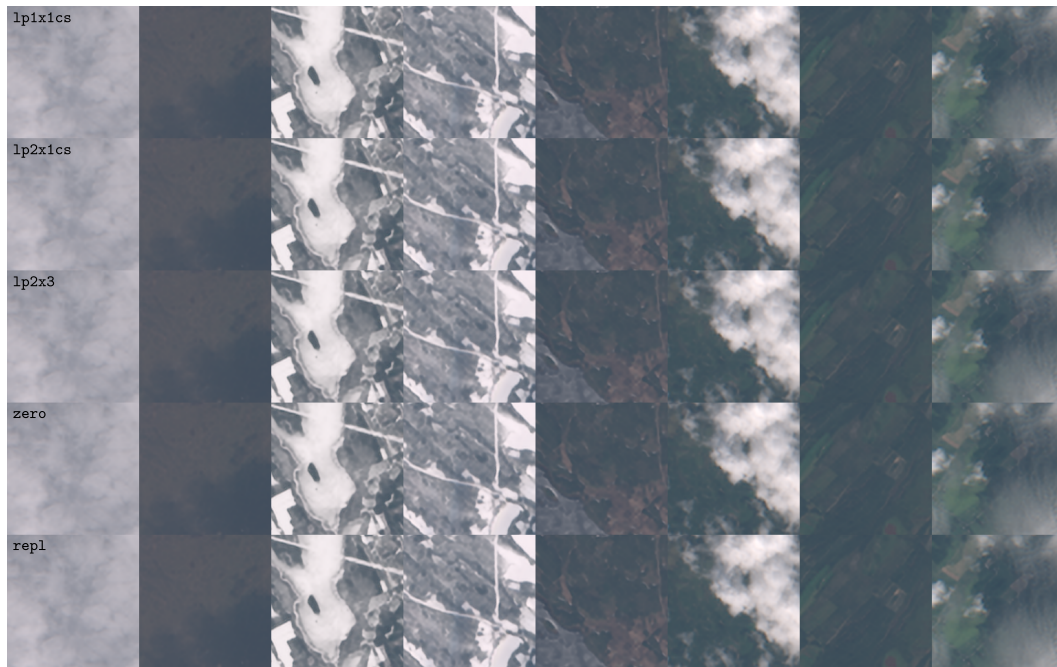


Figure 17: Stitched RVSR super-resolution results with output crop 1. Visually, the discontinuity artifact is much weaker than with output crop 1 and is only visible on high-contrast edges not perpendicular to the tile boundary.

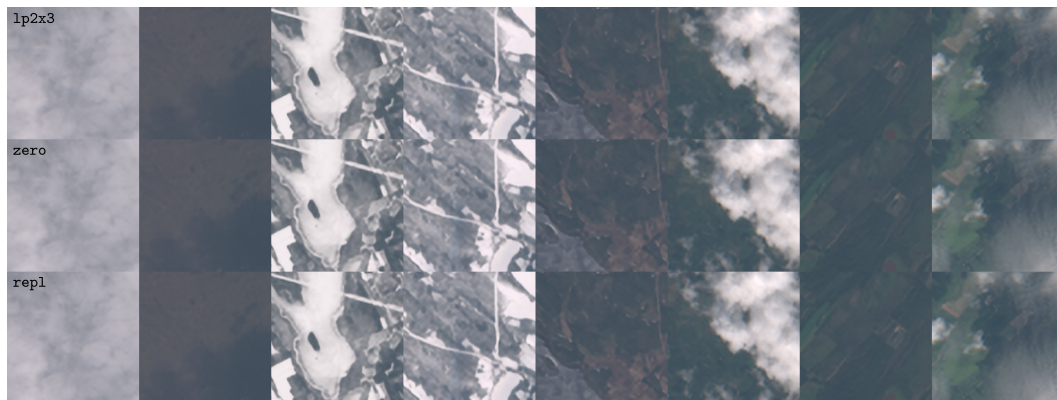


Figure 18: Stitched RVSR super-resolution results with output crop 5. No visible tiling artifacts.

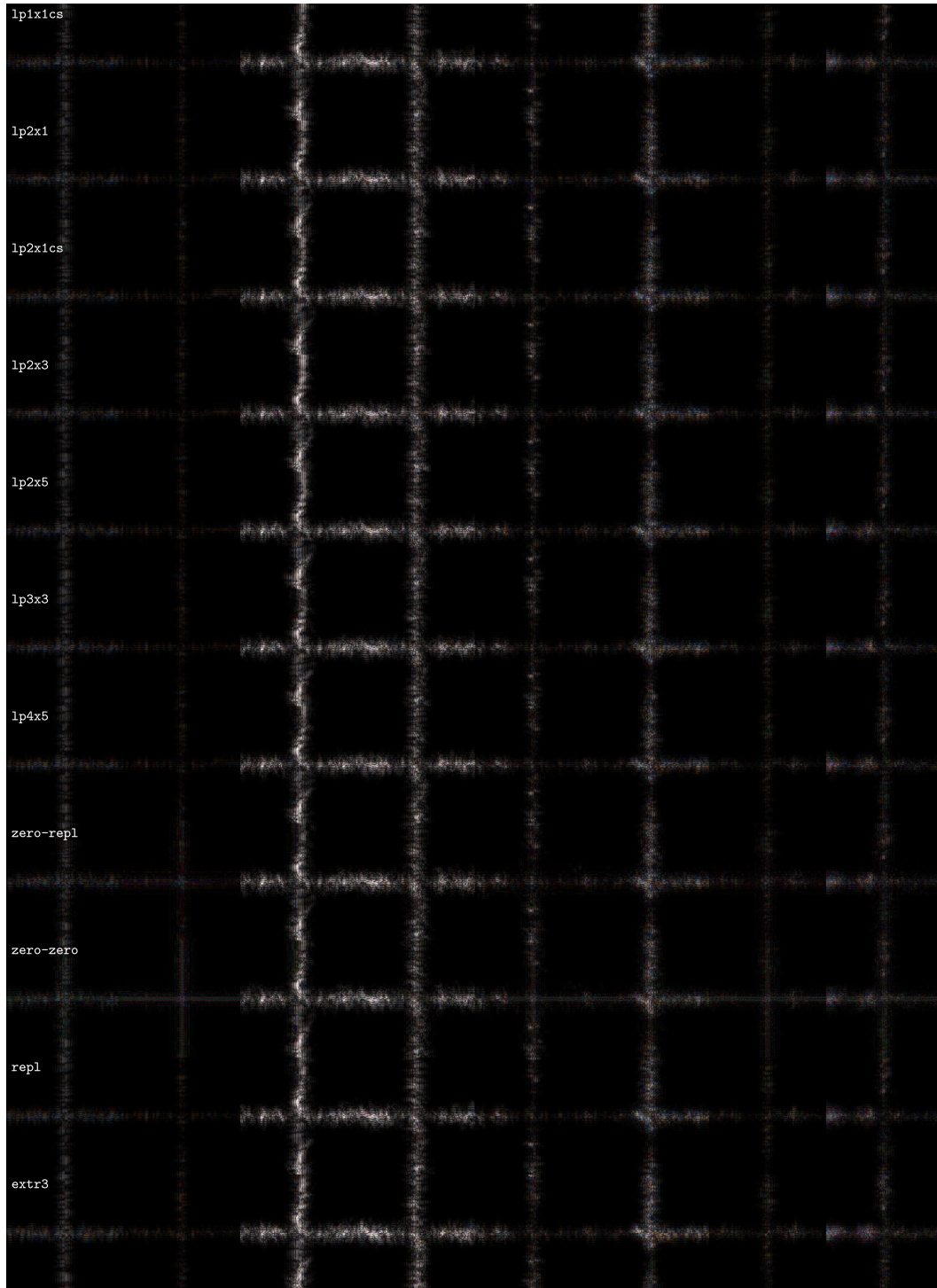


Figure 19: RVSr super-resolution deviation from shift equivariance for output crop 0, calculated as the absolute value of the difference between a stitched tiled prediction and a prediction using only valid convolutions. In these figures, black represents no difference and white represents an absolute reflectance difference of 0.2 on every channel.

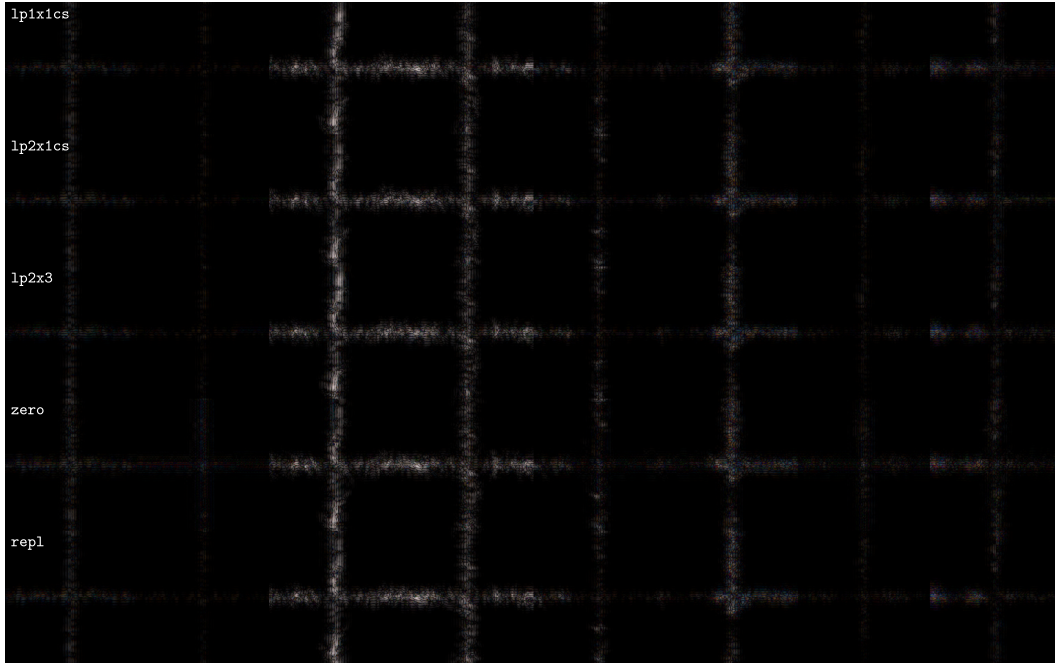


Figure 20: RVSr super-resolution deviation from shift equivariance for output crop 1.



Figure 21: RVSr super-resolution deviation from shift equivariance for output crop 5. zero displays elevated deviations compared to lp2x3 and repl.

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